Universidad Politécnica de Madrid

Escuela Técnica Superior de Ingenieros Informáticos

Máster Universitario en Digital Innovation

Master Thesis

Easy-to-Read (E2R) Adaptation of Figurative Language

Author: **Jelle van Lieshout**

Supervisor: **Dr. Mari Carmen Suárez–Figueroa**

Madrid, MM YYYY

Abstract

Figurative language, such as idioms and metaphors, presents a significant barrier to text accessibility and downstream Natural Language Processing tasks. While Large Language Models (LLMs) have shown promise in general language understanding, their ability to reliably handle figurative expressions in zero-shot scenarios remains under-explored compared to task-specific supervised models. This thesis investigates the effectiveness of agentic LLM pipelines for the detection, interpretation, and Easy-to-Read (E2R) adaptation of figurative language.

We propose a multi-stage agentic workflow that explicitly decomposes the task into detection, explanation, literal replacement, and self-verification. Using the SemEval-2022 Task 2 dataset for idioms and the VU Amsterdam Metaphor Corpus for metaphors, we evaluate the system's performance against both monolithic single-prompt LLM baselines and established supervised benchmarks.

We hypothesize that an agentic decomposition significantly improves the quality of literal replacements compared to standard prompting, particularly for complex metaphors that rely on abstract source-target mappings. Furthermore, we explore the trade-offs between semantic fidelity and readability in the generated replacements. This work aims to demonstrate that structured, observable agentic reasoning enables reliable figurative language interpretation and transformation without the need for extensive task-specific training, thereby contributing to more accessible and inclusive digital content.

Resumen

...

Acknowledgement

...

Contents

Abstract **A**

Resumen **B**

Acknowledgement **C**

1 Introduction **1**

2 Background & Related Work **2**
2.1 Figurative Language . . . 2
2.2 Datasets . . . 2
2.2.1 VU Amsterdam Metaphor Corpus (VUAMC) . . . 2
2.2.2 SemEval-2022 Task 2: Multilingual Idiomaticity Detection and Sentence Embedding . . . 2
2.2.3 Supporting Datasets (Planned) . . . 3
2.3 Evaluation Methodology . . . 3
2.3.1 Automatic Metrics . . . 3
2.3.2 Human Evaluation . . . 3
2.3.3 Multi-Layer Evaluation Framework . . . 4
2.4 Large Language Models & Agentic Systems . . . 4
2.5 Literature Review . . . 4

3 System Design & Methodology **6**
3.1 Problem Formulation . . . 6
3.2 Agentic Pipeline . . . 6
3.3 Infrastructure . . . 7
3.4 Evaluation Framework . . . 7
3.5 Baselines . . . 8

4 RQ1: Detection & Interpretation **9**
4.1 Experimental Setup . . . 9
4.2 Results . . . 9
4.3 Analysis . . . 10
4.4 Qualitative Examples . . . 10

5 RQ2: Agentic Decomposition **11**
5.1 Experiments . . . 11
5.2 Metrics . . . 11

6 RQ3: Replacement Quality **12**
6.1 Experimental Setup . . . 12
6.2 Idiom Replacement (SemEval 2022) . . . 12
6.3 Metaphor Replacement (VU Amsterdam) . . . 12
6.4 Analysis . . . 13

7 RQ4: Observability & Error Analysis **14**

8 Discussion & Limitations **15**

9 Conclusion & Future Work **16**

10 Annex **19**

List of Tables

Table 1 Zero-shot detection results (200 examples, `google/gemini-3-flash-preview`, $T = 0$). Token-level metrics are not reported for SemEval, as its gold annotations are span- and sentence-level only. Span precision and recall are available in Table 2. 9
Table 2 Span-level detection precision, recall, and F1 (IoU ≥ 0.5, macro-averaged). 9
Table 3 Idiom detect-then-replace results on SemEval 2022. BLEU and BERTScore are computed against gold Task B paraphrases. F1-sent and F1-span measure detection quality on the same examples. Note that the v1 and v2 sample sizes differ (100 vs. 20), so direct comparison carries variance. 12
Table 4 Metaphor detection metrics for detection-only vs. detect-then-replace runs (VU Amsterdam, prompt v1). The lower performance on the detect-then-replace task is partly attributable to the smaller sample (100 vs. 200 examples) and partly to the additional cognitive load imposed by simultaneous replacement. . . 13

List of Figures

List of Equations

1 Introduction

- Motivation: figurative language as a barrier to accessibility, simplification, and downstream NLP
- Limitations of supervised, task-specific approaches
- Rise of LLMs and agentic reasoning
- Contributions
  - Zero/one-shot/few-shot agentic system for figurative language handling
  - Unified treatment of idioms and metaphors (is this the case?)
  - Empirical evaluation of replacement quality [1]

2 Background & Related Work

2.1 Figurative Language

- Idioms vs metaphors (linguistic + cognitive distinction)
- Prior work on detection, classification, and simplification

2.2 Datasets

2.2.1 VU Amsterdam Metaphor Corpus (VUAMC)

The VU Amsterdam Metaphor Corpus [12] is a large, manually annotated corpus of English texts drawn from four genres: academic prose, news, fiction, and conversation. Metaphor annotation follows the MIP (Metaphor Identification Procedure) [13]: for each content word, annotators assess whether its contextual meaning contrasts with a more basic, concrete meaning from which the contextual meaning can be derived by comparison. Words meeting this criterion are labelled metaphorical.

The corpus is distributed as an XML file and contains 16,202 English sentences, each with per-token metaphoricity labels (binary: metaphorical / non-metaphorical). Crucially, the VUAMC provides *token-level* annotations rather than span- or sentence-level ones, making it well-suited for evaluating fine-grained detection but challenging for span-based models, since a single conceptual metaphor may involve multiple individually annotated tokens.

A custom Python client was developed to parse the XML, extract sentence-token structures, and derive character-offset spans from the token annotations. All 16,202 sentences are ingested into the experiment database. The VUAMC does *not* include gold literal paraphrases; replacement evaluation on this dataset therefore relies on detection metrics and human evaluation only.

2.2.2 SemEval-2022 Task 2: Multilingual Idiomaticity Detection and Sentence Embedding

SemEval-2022 Task 2 [16] is a shared task on idiom detection and interpretation in English, Portuguese, and Galician. This thesis uses the English (EN) partition of Subtask A and Subtask B:

- **Subtask A** provides sentences containing a pre-identified multi-word expression (MWE) and asks whether the MWE is used idiomatically in context (binary classification). Gold annotations include character-offset span boundaries for the MWE.
- **Subtask B** provides gold literal paraphrases of the full sentence for a subset of Subtask A examples, enabling evaluation of replacement quality.

The English partition contains 3,487 sentences after filtering and joining Tasks A and B. Gold paraphrases from Task B are stored alongside each example and used as references for BLEU and BERTScore evaluation. Unlike the VUAMC, SemEval 2022 Task 2 is evaluated at the sentence and span level, with no per-token labels.

2.2.3 Supporting Datasets (Planned)

- MAGPIE [4]: idiom interpretation robustness (qualitative / stress testing)
- TroFi [2]: verb metaphor detection (focused evaluation)
- MOH / MOH-X [9]: verb metaphor detection (secondary validation)
- MetaNet: conceptual metaphor grounding (interpretation support)

2.3 Evaluation Methodology

Standard automatic metrics are commonly applied to generation and paraphrase tasks, but their suitability for figurative language is limited. This section outlines the key metrics used in this thesis and discusses their limitations in the context of figurative-to-literal adaptation.

2.3.1 Automatic Metrics

BLEU. BLEU measures n-gram overlap between a system output and one or more reference texts [11]. It is precision-oriented and applies a brevity penalty. For figurative language replacement, BLEU is severely misaligned: literal paraphrases of idiomatic expressions intentionally avoid lexical overlap with the gold reference, and multiple semantically equivalent paraphrases may share no surface n-grams. BLEU is retained here primarily as a baseline point of comparison, but its low values should not be interpreted as evidence of poor replacement quality.

SARI. SARI decomposes quality into Add, Delete, and Keep operations relative to the input and references, and was originally designed for text simplification [17]. Figurative-to-literal transformation closely resembles simplification (replacing opaque expressions with transparent ones), making SARI conceptually well-motivated. Its practical limitation is reference dependence; it is not yet implemented but is under consideration.

BERTScore. BERTScore computes contextual token-embedding similarity between generated and reference texts using a pre-trained language model (RoBERTa in this work), enabling soft alignment rather than exact string matching [18]. This makes it more robust to paraphrase variation than BLEU. However, high BERTScore does not guarantee correct interpretation: a plausible but semantically incorrect paraphrase may still achieve a high score. BERTScore is used as the primary automatic metric for replacement evaluation.

2.3.2 Human Evaluation

Automatic metrics alone are insufficient for assessing replacement quality. Human evaluation is planned for a sample of 40–50 sentences using a *meaning fidelity* scale (1–5):

- 1 — Incorrect or misleading interpretation;
- 3 — Partially correct (captures some but not all of the intended meaning);
- 5 — Fully meaning-preserving literal restatement.

Annotators will also be asked to provide their own literal paraphrase when the system's output is rated below 4, enabling qualitative error analysis.

2.3.3 Multi-Layer Evaluation Framework

The evaluation protocol is structured as four complementary layers, inspired by the framework proposed in [14]:

1. **Detection accuracy**: span-level precision, recall, and F1 (Section 3.4);
2. **Semantic similarity**: BERTScore and BLEU (supporting/automatic);
3. **Interpretive correctness**: human evaluation of meaning fidelity (primary);
4. **Conceptual grounding**: correctness of source–target metaphor mappings and idiom sense disambiguation (planned).

2.4 Large Language Models & Agentic Systems

- Zero-shot, one-shot, and few-shot prompting
- Agentic decomposition (planning, reflection, self-verification)
- Observability (LangSmith-style traces)

2.5 Literature Review

Relevant papers to include:

- [14] Mari Carmen Suárez-Figueroa, Isam Diab, Edna Ruckhaus, and Isabel Cano. First steps in the development of a support application for easy-to-read adaptation. *Universal Access in the Information Society*, 23:365–377, 3 2024
- [15] Mari Carmen Suárez-Figueroa, Isam Diab, Álvaro González, and Jesica Rivero-Espinosa. Towards an automatic easy-to-read adaptation of morphological features in spanish texts. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 14142 LNCS:176–198, 2023
- [3] Alexander Vladislavovitch Dmitrijev, Elena Sergeevna Krupnova, and Anastasia Aleksandrovna Protopopova. Metaphors and analogies in the context of large language models. In *Scenarios, Fictions, and Imagined Possibilities in Science, Engineering, and Education*, volume 1203 LNNS, pages 326–341. Springer Science and Business Media Deutschland GmbH, 2024
- [10] Arthur Neidlein, Philipp Wiesenbach, and Katja Markert. An analysis of language models for metaphor recognition. *COLING 2020 - 28th International Conference on Computational Linguistics, Proceedings of the Conference*, pages 3722–3736, 2020
- [8] Emmy Liu, Chenxuan Cui, Kenneth Zheng, and Graham Neubig. Testing the ability of language models to interpret figurative language. *NAACL 2022 - 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Proceedings of the Conference*, pages 4437–4452, 4 2022
- [7] Huiyuan Lai and Malvina Nissim. A survey on automatic generation of figurative language: From rule-based systems to large language models. *ACM Computing Surveys*, 56, 5 2024

- [5] Nicholas Ichien, Dušan Stamenković, and Keith J. Holyoak. Large language model displays emergent ability to interpret novel literary metaphors. *Metaphor and Symbol*, 39:296–309, 10 2024

- [6] Kaidi Jia, Yanxia Wu, Ming Liu, and Rongsheng Li. Curriculum-style data augmentation for llm-based metaphor detection. arXiv:2412.02956 [cs.CL], 12 2024

3 System Design & Methodology

3.1 Problem Formulation

The core task addressed by this thesis is figurative language adaptation: given a raw English sentence that may contain figurative expressions (idioms or metaphors), produce a literal, meaning-preserving restatement alongside explicit detection annotations. More formally, the system accepts a sentence s and outputs:

- A set of detected figurative spans $\{(e_i, \text{start}_i, \text{end}_i)\}$, where each e_i is a verbatim copy of an expression from s with character-offset boundaries;
- A per-token binary label sequence $\mathbf{l} \in \{0, 1\}^{|\mathcal{T}(s)|}$ over the whitespace-tokenised form of s, derived from the detected spans;
- A brief linguistic explanation of the figurative usage;
- Optionally, a literal paraphrase $\hat{s}$ of the full sentence, with all figurative expressions replaced by meaning-preserving literal alternatives.

Two task variants are supported: *detection* (spans, labels, and explanation only) and *detect-then-replace* (additionally producing $\hat{s}$). Both are evaluated over two phenomena independently: *metaphors*, benchmarked against the VU Amsterdam Metaphor Corpus [12], and *idioms*, benchmarked against SemEval 2022 Task 2 [16]. A notable challenge is that some metaphors encode meaning that resists full literalisation without loss; replacement decisions therefore require nuanced linguistic understanding beyond surface paraphrase.

3.2 Agentic Pipeline

The system is implemented as four LangGraph workflows, one per (phenomenon, task) combination:

- `metaphor_detection_graph` — detects metaphorical words or phrases, following VUAMC MIP (Metaphor Identification Procedure) guidelines;
- `idiom_detection_graph` — detects multi-word idiomatic expressions, following SemEval 2022 Task 2 annotation conventions;
- `metaphor_detect_then_replace_graph` — detects metaphors and produces a literal sentence paraphrase;
- `idiom_detect_then_replace_graph` — detects idioms and produces a literal sentence paraphrase.

Each workflow issues a single structured LLM call whose output must conform to the `TaskOutput` Pydantic schema:

- `detection.is_figurative` (boolean): whether the sentence contains any figurative expression;
- `detection.figurative_expressions` (list of strings): verbatim copies of each detected expression, preserving casing and punctuation as they appear in the input text;

- `detection.explanation` (string): brief linguistic reasoning for the detection decision;
- `replacement.literal_paraphrase` (optional string): the complete sentence rewritten with all figurative expressions replaced.

Structured output is enforced via LangChain's `with_structured_output()` interface, which uses JSON-schema-constrained decoding or function-calling depending on the model's capabilities. Outputs violating the schema trigger automatic retries.

Two deterministic post-processing steps follow the LLM call:

1. **Span extraction**: each verbatim expression in `figurative_expressions` is located in the input sentence by exact string search, with a case-insensitive fallback. Matched positions are converted to `Span(start, end)` objects with inclusive-start, exclusive-end character offsets. Overlapping spans are merged; all spans are clipped to the sentence boundary.
2. **Token label derivation**: a binary label is assigned to each whitespace-delimited token by testing whether any detected span overlaps the token's character range. This produces a label sequence directly comparable to VUAMC's token-level metaphor annotations.

3.3 Infrastructure

The system is built on the following components:

- **Backend**: FastAPI (Python 3.13) on port 3030. Experiment runs are launched asynchronously as background tasks, enabling concurrent execution.
- **LLM access**: OpenRouter API client wrapping LangChain's `ChatOpenAI`. The primary model used for all experiments reported here is `google/gemini-3-flash-preview`, selected for its reliable structured-output compliance. Free-tier alternatives (e.g. Gemma 3 12B, Mistral Small 24B) are available for exploratory runs.
- **Persistence**: Couchbase 7.6 document store with four collections: `datasets` (gold-annotated examples; 16,202 VU Amsterdam + 3,487 SemEval = 19,689 total), `runs` (experiment metadata and status), `predictions` (per-example LLM outputs), and `evaluations` (computed metrics). Document keys follow a namespaced convention (e.g. `prediction::{run_id}::{e`
- **Prompt versioning**: each run records the full prompt text and its SHA-256 hash, enabling exact reproducibility and detection of prompt drift between runs.
- **Observability**: LangSmith tracing is configured for all LLM invocations, providing post-hoc inspection of model inputs, outputs, token usage, and latency per step. Trace analysis is deferred to Chapter **??**.
- **Frontend**: a React/TypeScript single-page application with TanStack Query for data fetching, featuring run dashboards, example-level span visualisation with colour-coded gold/predicted overlays, literal replacement diff views, and multi-run metric comparison charts.

3.4 Evaluation Framework

Five evaluation metrics are computed automatically for each completed run:

- **Token F1**: binary precision, recall, and F1 aggregated across all token labels within a run, computed using scikit-learn's `precision_recall_fscore_support`. Applicable only when gold token-level annotations are available (VU Amsterdam).
- **Span F1**: macro-averaged precision, recall, and F1 over detected character-offset spans. A predicted span is a true positive if it overlaps any unmatched gold span with Intersection-over-Union (IoU) ≥ 0.5, enforcing one-to-one matching.
- **Sentence F1**: binary classification F1, where a sentence is labelled figurative if and only if at least one span is detected, for both gold and predicted.
- **BLEU**: sentence-level BLEU with NLTK smoothing (method 1) against the gold paraphrase from SemEval Task B. Applicable only to detect-then-replace runs on SemEval.
- **BERTScore** [18]: contextual embedding similarity (RoBERTa model) between predicted and gold replacement sentences, returning precision, recall, and F1. Applicable only to detect-then-replace runs on SemEval.

All metrics are persisted as `EvaluationData` Couchbase documents and are re-computable on demand via the `POST /runs/{run_id}/evaluate` endpoint. Section 2.3 discusses the appropriateness of these metrics for figurative language evaluation in more detail.

3.5 Baselines

- *Monolithic single-prompt* (to be implemented): a single prompt that requests detection and replacement without intermediate explanation or structured reasoning.
- *Detection-only comparison*: detect-then-replace runs are compared against detection-only runs on the same examples to quantify the effect of the replacement step on detection quality (see Section 6).
- *Naïve paraphrasing* (to be implemented): a baseline that rewrites the sentence without explicit figurative language handling.

4 RQ1: Detection & Interpretation

To what extent can a zero-/one-shot agentic LLM pipeline accurately detect and interpret idiomatic and metaphorical expressions in context?

4.1 Experimental Setup

All detection experiments use the following configuration:

- **Model**: `google/gemini-3-flash-preview` via OpenRouter;
- **Temperature**: 0 (greedy decoding, for reproducibility);
- **Prompt version**: v1 (zero-shot, no few-shot examples);
- **Subset size**: 200 examples per dataset, sampled from the beginning of each ingested collection.

For metaphor detection, the prompt instructs the model to follow the MIP procedure: identify each word whose contextual meaning contrasts with its most basic, concrete meaning, and return those words verbatim as `figurative_expressions`. For idiom detection, the prompt instructs the model to identify multi-word expressions whose combined meaning cannot be derived compositionally.

In both cases, the LLM output is post-processed to extract character-offset spans and derive per-token binary labels (Section 3.4). These are compared against gold annotations from the respective datasets.

4.2 Results

Table 1 summarises the detection results for both datasets.

Dataset	Task	F1-sent	F1-tok	P-tok	R-tok	F1-span
VU Amsterdam	Metaphor detection	0.452	0.441	0.460	0.424	0.202
SemEval 2022	Idiom detection	0.655	—	—	—	0.389

Table 1: Zero-shot detection results (200 examples, `google/gemini-3-flash-preview`, $T = 0$). Token-level metrics are not reported for SemEval, as its gold annotations are span- and sentence-level only. Span precision and recall are available in Table 2.

Dataset	Task	P-span	R-span	F1-span
VU Amsterdam	Metaphor detection	0.244	0.191	0.202
SemEval 2022	Idiom detection	0.369	0.430	0.389

Table 2: Span-level detection precision, recall, and F1 (IoU ≥ 0.5, macro-averaged).

4.3 Analysis

Idiom detection outperforms metaphor detection at all levels. Sentence-level F1 is 0.655 for idioms vs. 0.452 for metaphors, and span-level F1 is 0.389 vs. 0.202. This is consistent with the structural properties of each phenomenon: idioms are discrete, multi-word expressions with clearly delimited boundaries (e.g. *keep an eye on*, *spill the beans*), making them more amenable to verbatim extraction and span matching. Metaphors, by contrast, are annotated at the individual token level in the VUAMC, often involving isolated words whose figurative use depends on broader context. A single conceptual metaphor may manifest as multiple separately annotated tokens, making exact span recovery harder.

Span F1 is lower than sentence and token F1 for metaphors. The span IoU ≥ 0.5 threshold is sensitive to the mismatch between character offsets derived from LLM verbatim outputs and the gold token boundaries in the VUAMC. A word correctly identified as metaphorical but copied with a slight boundary offset (e.g. including punctuation) will fail the IoU threshold. Token F1 (0.441) and sentence F1 (0.452) are therefore more informative metrics for this dataset.

Comparison to supervised benchmarks. A direct comparison against supervised models reported in the VUAMC and SemEval literature is planned for this chapter. Initial observations suggest that the zero-shot pipeline achieves competitive sentence-level performance on idiom detection, while metaphor token F1 (0.441) is below the best reported supervised results but above naive baselines.

4.4 Qualitative Examples

5 RQ2: Agentic Decomposition

Does agentic decomposition improve figurative language handling compared to monolithic prompting?

5.1 Experiments

- Same inputs, different architectures
- Ablation:
  - no explanation step
  - no self-verification
  - full agentic pipeline

5.2 Metrics

- Replacement quality (see Chapter 6)
- Error types
- Failure traceability

6 RQ3: Replacement Quality

How effectively can figurative expressions be replaced with literal, meaning-preserving alternatives?

6.1 Experimental Setup

Replacement experiments use the *detect-then-replace* task variant: the model detects figurative spans and in the same call produces a literal paraphrase of the full sentence. All experiments use `google/gemini-3-flash-preview`, temperature 0. Two prompt versions are evaluated:

- **Prompt v1**: a minimal stub instructing the model to identify figurative expressions and provide a literal paraphrase (three lines of instruction).
- **Prompt v2**: a chain-of-thought prompt instructing the model to (1) identify each expression, (2) reason about its meaning in context, and (3) rewrite the full sentence with all figurative parts replaced, keeping non-figurative content unchanged. Includes one illustrative example.

6.2 Idiom Replacement (SemEval 2022)

SemEval 2022 Task B provides gold literal paraphrases for a subset of Task A sentences, enabling automatic evaluation of replacement quality using BLEU and BERTScore. Table 3 reports results for two runs: 100 examples with prompt v1 and 20 examples with prompt v2.

Prompt	N	F1-sent	F1-span	BLEU	BERTScore F1
v1 (stub)	100	0.750	0.352	0.047	0.847
v2 (chain-of-thought)	20	0.600	0.200	0.158	0.863

Table 3: Idiom detect-then-replace results on SemEval 2022. BLEU and BERTScore are computed against gold Task B paraphrases. F1-sent and F1-span measure detection quality on the same examples. Note that the v1 and v2 sample sizes differ (100 vs. 20), so direct comparison carries variance.

The prompt v2 improves replacement quality substantially: BLEU increases from 0.047 to 0.158 and BERTScore F1 from 0.847 to 0.863. The slight drop in sentence and span F1 for v2 is likely due to sample variance (20 examples). These results suggest that chain-of-thought prompting with an explicit reasoning step improves literal paraphrase quality even in a zero-shot setting.

BLEU values are low overall (0.047–0.158), which is expected given the nature of the task: valid literal paraphrases of idiomatic expressions often share no n-grams with the gold reference. For example, the idiom *smoking gun* may be correctly replaced with *decisive evidence* or *conclusive proof*, neither of which overlaps with the reference. BERTScore (0.847–0.863) provides a more informative signal, capturing semantic proximity at the embedding level.

6.3 Metaphor Replacement (VU Amsterdam)

The VUAMC does not include gold literal paraphrases, making BLEU and BERTScore inapplicable. Replacement quality for metaphors must therefore be assessed through (a) detection

metrics on detect-then-replace runs and (b) human evaluation (planned). Table 4 compares detection metrics for detection-only vs. detect-then-replace runs on the same dataset.

Task	**N**	**F1-sent**	**F1-tok**	**F1-span**
Detection only	200	0.452	0.441	0.202
Detect-then-replace	100	0.293	0.318	0.044

Table 4: Metaphor detection metrics for detection-only vs. detect-then-replace runs (VU Amsterdam, prompt v1). The lower performance on the detect-then-replace task is partly attributable to the smaller sample (100 vs. 200 examples) and partly to the additional cognitive load imposed by simultaneous replacement.

Detection metrics are lower on the detect-then-replace task, suggesting that asking the model to simultaneously produce a replacement paraphrase may shift attention away from precise span localisation. This is consistent with observations in the SemEval results (lower F1-span for v1 replacement vs. detection-only).

6.4 Analysis

BLEU is poorly suited to replacement evaluation. Even for correctly paraphrased idioms, BLEU scores are in the range 0.05–0.16, confirming the known limitations of n-gram overlap metrics for tasks with multiple valid outputs. BERTScore is the preferred automatic metric; human evaluation is required for a definitive assessment.

Idioms are replaced more reliably than metaphors. This is consistent with Hypothesis H3: idiomatic expressions have more constrained and conventionalised meanings, making them more tractable targets for zero-shot paraphrase generation. Metaphors, particularly those involving abstract source–target mappings, require a deeper understanding of the specific context, which current zero-shot prompting may not consistently achieve.

Replacement degrades detection slightly. Adding the replacement requirement reduces detection F1 on both datasets. For future experiments, it may be worth separating detection and replacement into two sequential LLM calls to avoid this interference.

Human evaluation (planned). An annotation study covering 40–50 examples from the detect-then-replace runs is planned, using the meaning fidelity scale described in Section 2.3. Annotators will rate the system's literal paraphrase on a 1–5 scale and optionally provide a preferred alternative. Examples for annotation will be selected using the `GET /runs/{run_id}/export` endpoint, filtering for shorter sentences ($<$ 20 words) where the system produced a non-empty replacement.

7 RQ4: Observability & Error Analysis

How observable and debuggable are agentic systems compared to end-to-end prompting?

- Case studies using LangSmith traces
- Error localization
- Correlation between explanation quality and outcome quality

8 Discussion & Limitations

- What agentic systems can/cannot do
- Where metaphors fundamentally resist literalization
- Limitations:
  - Conceptual: metaphors encoding meaning that cannot be fully literalized; replacement oversimplifying nuance
  - Evaluation limits: Subjectivity of human evaluation; no gold standard for metaphor replacement
  - Model dependence: Results vary across LLM providers
  - Scope limits: Sentence/MWE level focus; no discourse-wide tracking; primarily English

9 Conclusion & Future Work

- Summary of findings
- Implications for NLP accessibility
- Directions: multilinguality, discourse-level metaphors, human-in-the-loop

References

[1] A. Author. Sample article title. *Journal Name*, 1(1):1–10, 2024.

[2] Julia Birke and Anoop Sarkar. A clustering approach for the nearly unsupervised recognition of nonliteral language. In *Proceedings of the 11th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, pages 329–336, 2006.

[3] Alexander Vladislavovitch Dmitrijev, Elena Sergeevna Krupnova, and Anastasia Aleksandrovna Protopopova. Metaphors and analogies in the context of large language models. In *Scenarios, Fictions, and Imagined Possibilities in Science, Engineering, and Education*, volume 1203 LNNS, pages 326–341. Springer Science and Business Media Deutschland GmbH, 2024.

[4] Hessel Haagsma, Johan Bos, and Malvina Nissim. MAGPIE: A large corpus of potentially idiomatic expressions. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*, pages 279–287. European Language Resources Association, 2020.

[5] Nicholas Ichien, Dušan Stamenković, and Keith J. Holyoak. Large language model displays emergent ability to interpret novel literary metaphors. *Metaphor and Symbol*, 39:296–309, 10 2024.

[6] Kaidi Jia, Yanxia Wu, Ming Liu, and Rongsheng Li. Curriculum-style data augmentation for llm-based metaphor detection. arXiv:2412.02956 [cs.CL], 12 2024.

[7] Huiyuan Lai and Malvina Nissim. A survey on automatic generation of figurative language: From rule-based systems to large language models. *ACM Computing Surveys*, 56, 5 2024.

[8] Emmy Liu, Chenxuan Cui, Kenneth Zheng, and Graham Neubig. Testing the ability of language models to interpret figurative language. *NAACL 2022 - 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Proceedings of the Conference*, pages 4437–4452, 4 2022.

[9] Saif Mohammad, Ekaterina Shutova, and Peter Turney. Metaphor as a medium for emotion: An empirical study. In *Proceedings of the 5th Joint Conference on Lexical and Computational Semantics*, pages 23–33. Association for Computational Linguistics, 2016.

[10] Arthur Neidlein, Philipp Wiesenbach, and Katja Markert. An analysis of language models for metaphor recognition. *COLING 2020 - 28th International Conference on Computational Linguistics, Proceedings of the Conference*, pages 3722–3736, 2020.

[11] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. BLEU: a method for automatic evaluation of machine translation. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*, pages 311–318. Association for Computational Linguistics, 2002.

[12] Gerard J Steen, Aletta G Dorst, J Berenike Herrmann, Anna A Kaal, and Tina Krennmayr. VU amsterdam metaphor corpus, 2010. Oxford Text Archive.

[13] Gerard J. Steen, Aletta G. Dorst, J. Berenike Herrmann, Anna A. Kaal, Tina Krennmayr, and Trijntje Pasma. *A Method for Linguistic Metaphor Identification: From MIP to MIPVU*. John Benjamins Publishing Company, Amsterdam, 2010.

[14] Mari Carmen Suárez-Figueroa, Isam Diab, Edna Ruckhaus, and Isabel Cano. First steps in the development of a support application for easy-to-read adaptation. *Universal Access in the Information Society*, 23:365–377, 3 2024.

[15] Mari Carmen Suárez-Figueroa, Isam Diab, Álvaro González, and Jesica Rivero-Espinosa. Towards an automatic easy-to-read adaptation of morphological features in spanish texts. *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, 14142 LNCS:176–198, 2023.

[16] Harish Tayyar Madabushi, Edward Gow-Smith, Marcos Garcia, Carolina Scarton, Marco Idiart, and Aline Villavicencio. SemEval-2022 task 2: Multilingual idiomaticity detection and sentence embedding. In Guy Emerson, Natalie Schluter, Gabriel Stanovsky, Ritesh Kumar, Alexis Palmer, Nathan Schneider, Siddharth Singh, and Shyam Ratan, editors, *Proceedings of the 16th International Workshop on Semantic Evaluation (SemEval-2022)*, pages 107–121, Seattle, United States, July 2022. Association for Computational Linguistics.

[17] Wei Xu, Courtney Napoles, Ellie Pavlick, Quanze Chen, and Chris Callison-Burch. Optimizing statistical machine translation for text simplification. In *Transactions of the Association for Computational Linguistics*, volume 4, pages 401–415, 2016.

[18] Tianyi Zhang, Varsha Kishore, Felix Wu, Kilian Q. Weinberger, and Yoav Artzi. BERTScore: Evaluating text generation with BERT. In *International Conference on Learning Representations*, 2020.

10 Annex

...